Test the complete system and the source shift

New public-source evidence and full-stream comparisons

Completed in this pilot

740 public action clips acquired
36 CLIP classifier fits
Original / LoRA SmolVLM transfer
10 complete streams, 3 detector arms

What the results establish

Tiling worsens primary event detection
SmolVLM LoRA regresses proxy AP
Source-specific CLIP results follow
No automatic model promotion

More data and better box recall do not guarantee better target-camera performance.

Public data acquired, with explicit label limits

XRF55 release v6 • smoking action + drinking / eating / phone controls

Split	Scene blocks	Subject aliases	Clips	Positive / control
Train	1 + 2	15	340	85 / 255
Development	3	2	160	40 / 120
Test	4	3	240	60 / 180

Masked training actors: treat this as a scripted gesture proxy; cigarettes were not verified.

Same selection rules across old, new and mixed data

New public-source evidence and full-stream comparisons

Thirty-six classifier fits

2 frozen pretrained CLIP encoders
3 training-source families
2 full / contextual-crop recipes
C = 0.1, 1, 10; group-balanced fits

Development-only decisions

Same 1,474 dev cases / 4 sources
One cutoff: each source FPR ≤5%
Select by worst-source recall
Freeze both models before test

Old-only and new-only differ in training data; both receive combined-source dev supervision.

MobileCLIP: data addition has source-specific effects

Frozen every-source development cutoff • CCTV n=646; proxy n=240

Held-out source	Training family	AP	Recall	FPR
Historical CCTV	old only	0.6937	10.2%	5.5%
Historical CCTV	new only	0.6104	0.0%	0.0%
Historical CCTV	mixed	0.5673	12.2%	15.8%
New action proxy	old only	0.4978	0.0%	0.0%
New action proxy	new only	0.4553	0.0%	0.0%
New action proxy	mixed	0.5028	0.0%	11.1%

AP is ranking. Recall and FPR measure each family's frozen operating point.

SigLIP2: data addition has source-specific effects

Frozen every-source development cutoff • CCTV n=646; proxy n=240

Held-out source	Training family	AP	Recall	FPR
Historical CCTV	old only	0.7309	8.9%	4.0%
Historical CCTV	new only	0.5631	0.0%	0.4%
Historical CCTV	mixed	0.6227	36.9%	36.8%
New action proxy	old only	0.3568	0.0%	0.0%
New action proxy	new only	0.6313	61.7%	15.6%
New action proxy	mixed	0.7768	21.7%	1.7%

AP is ranking. Recall and FPR measure each family's frozen operating point.

SmolVLM: prior LoRA does not improve this source

Matched original / prior adapter • eight views each • no new training

Route	Cutoff calibration	AP	Recall	FPR
original	New source dev	0.3855	61.7%	36.7%
original	19 Oct dev	0.3855	0.0%	0.0%
lora	New source dev	0.3573	35.0%	18.3%
lora	19 Oct dev	0.3573	0.0%	0.0%

Both older budget cutoffs miss all positives; new-source calibration still exceeds 5% test FPR.

Full-stream tiling reverses the component gain

AVSS EASY / MEDIUM primary • unchanged detector, tracker and alarm rules

Detector arm	Events found	Normal FP	Duplicates	Median delay
full	2/2	0	0	34.82 s
item tiles	1/2	1	0	35.74 s
all tiles	1/2	1	1	13.25 s

Retain full-frame baseline. Better CAVIAR box recall did not improve the event pipeline.

Keep provisional and censored evidence separate

Seven ABODA streams share two scene blocks; AVSS is one conservative source block

Evidence	Arm	Found	FP	Repeat	Unknown
ABODA provisional	full	4/5	0	0	1
ABODA provisional	item tiles	2/5	0	0	5
ABODA provisional	all tiles	2/5	1	0	4
AVSS HARD censored	full	1/1	0	1	0
AVSS HARD censored	item tiles	1/1	0	1	0
AVSS HARD censored	all tiles	1/1	1	1	0

Ten recordings are not ten independent cameras. Unknown-time alarms remain unscored.

Replay and native forwards bind the results

New public-source evidence and full-stream comparisons

Numerical / data checks

7,536 stream decisions replayed
All 36 candidates and 6 choices
All selected / reference test scores
11,840 decoded new inputs checked

Native GPU validation

Both CLIP encoders, full + crop
Fixed train / dev / test cases
Original and LoRA SmolVLM paths
Source and code identities saved

Passed verification checks computation and identity; it does not strengthen proxy labels.

Next evidence needed for the five objectives

No target is promoted from this small public-source pilot

Objective	Next concrete data / modelling target
Smoking	Visible-cigarette CCTV evidence; distant drinking / eating / phone negatives
Unattended items	Owner/item histories; coherent association and gap recovery; long normal streams
Littering	Annotated release/disposal events and putting-down / picking-up controls
Loitering	Duration and zone-labelled person tracks; camera/day holdouts
Apparent public sleep	Temporal resting/wake controls and new locations; visual definition only

Prioritize independent event evidence and a frozen acceptance test over a larger parameter search.